

1 Overview

Generative AI is disrupting education and society. Today's computing students will need **professional skills** to continuously learn, co-design, develop, and lead generative AI integration in society as members of adaptive high-performing teams. For example, decades of organizational psychology research and industry research like Google's Rework study found the top skill enabling high-performing software teams is psychological safety [101]: choosing, in spite of potential reprisal or negative judgment, to make yourself vulnerable by raising difficult issues or asking for help. However, current approaches to generative AI (GenAI) in computing education focus primarily on teaching GenAI tool use without developing the critical reflection [128, 144] and professional skills students will need to continuously learn and shape this rapidly evolving landscape.

The gap in teaching professional skills in computing has actually persisted for decades. Review and aggregate studies identify continuous and lifelong learning [52, 56], and oral and written communication [112] as top skill deficits, even with helpful but insufficient approaches like team projects in courses [52, 110].

This gap persists because better approaches are rarely scalable or easy to adopt [61]. Faculty say more resources are needed to teach and give feedback on professional skills [135], but faculty are chronically under-resourced and hesitate to add more topics into already packed courses. Professional skills are also comparatively undervalued, so they hesitate to adopt or create a dedicated professional skills course.

We also lack research on effective ways for learning key professional skills, such as psychological safety and critical and ethical use of generative AI (GenAI). The pace of change is so rapid that computing education research is still figuring out how to teach the use of coding assistants like Co-Pilot, while professionals are learning to use coding agents through experimentation and reflection. Unless we find a systematic way to catch up, computing students may be left behind, based on growing post-graduation employment gaps.

This proposal seeks to fill the gap in scalable, adoptable, and effective techniques for learning professional skills in computing education, including lifelong learning and critical and reflective GenAI use, via T1) student reflection groups that work via peer learning with minimal instructor time, T2) video-assisted comparative reflection on GenAI use, and T3) building tools to connect students and professionals to reflect, improve, and learn together in mutually beneficial ways

I aim to advance knowledge in computing education research (CER) and human-computer interaction (HCI); this includes novel designs for and longitudinal mixed-methods empirical evaluations advancing our understanding for: 1) teaching and reflection methods for learning critical and ethical GenAI use, 2) video-assisted comparative reflection by learners inspired by qualitative research techniques, 3) empirical understanding of GenAI use, including critical reflection on use, learning over time, and how, when, and why students do and do not use GenAI, 4) supporting group reflection in synchronous online settings, and 5) bridging student and professional learning communities to reflect and improve together. This work will build foundations for improved reflection and lifelong learning practices that contribute to cultural change for more critical and ethical computing in general and for GenAI, across education and industry, enabled by stronger, more vulnerable human relationships and professional networks.

To do this, I will draw on my interdisciplinary research background in CER and HCI, including theory-based design and evaluation [65, 97, 100], validity and educational assessment [96, 98, 99], and building learning technology [100]. I have begun doing so in preliminary work that suggests video-assisted reflection aids learning critical GenAI use skills [47], and weekly outside-of-class non-project team reflection groups aid learning professional skills [95].

I will integrate this research program into my teaching and outreach. As a faculty member who teaches online, my plan includes developing an adoptable one credit reflection groups course on professional skills, a three credit 200-level course on professional skills including lifelong learning and GenAI, creating faculty development workshops for implementing reflection groups, organizing workshops and community events for students and faculty across institutions to discuss and co-design integrating critical and ethical GenAI

use into education, and building open-source curricular materials. I will also create a new outreach program inviting high school students to virtually attend and/or participate in reflection groups with student mentors.

1.1 Key Insights, Intellectual Merit, and Approach

Building on my expertise in interdisciplinary research, my approach leverages several key insights in a complementary way to address gaps in prior work, transforming the state of the art in reflection pedagogy for computing education and generative AI education:

1. **Scalability via novel designs for peer learning and mutually beneficial student-professional learning environments:** I will create novel designs to scaffold and thus leverage peer learning in student reflection groups for professional skills, and explore ways to augment groups with alumni and professionals. Individual reflection assignments where a student effectively gives themselves feedback are also scalable, with well-designed scaffolding. My key insights include combining group and individual reflection to complement and reinforce each other, and also adapting prior work from experimental psychology research [12] and professional medical and teacher training group reflection practices [105].
2. **Adoption via designing adoption pathways:** I will create a spectrum, from drop-in mostly outside-of-class reflection assignments, to whole course designs for professional skills. Instructors can implement one, see the benefits, then gradually adopt more, creating smoother adoption paths. My key insight is to design for adoption; for example, I will conduct co-design with faculty to understand and address adoption barriers in different types of institutions, and evaluate adoption as a part of design iteration. I will draw on implementation science [16], which studies adoption, and studies of adoption in software engineering [71, 104].
3. **Effectiveness via novel design for comparative individual reflection made easier with video assistance:** Video-assisted reflection reduces memory load and cognitive biases in recall. My key insight is to create novel designs for video-assisted *comparative* reflection. For example, learners will record videos of different programming sessions, varying different ways of using and not using GenAI, then reviewing and comparing them. I will also adapt ideas from research on self-experimentation.
4. **Effectiveness via design iteration using longitudinal evaluation in courses and after graduation:** I will design based on computing education and general education theories, evaluate professional skills longitudinally including in actual professional contexts, then iterate to improve effectiveness. The evaluations will also contribute to HCI for empirical understanding of GenAI use, including critical reflection on use, learning over time, and how, when, and why students do and do not use GenAI.

The proposed research approach has three interrelated but distinct research tasks:

Task 1) Reflection Groups for Professional Skills Development: Building on my successful pilot, I will iteratively design, implement, and evaluate peer reflection groups that support developing professional skills including iterative improvement, lifelong learning, psychological safety, and communication. Unlike project teams, these groups focus specifically on supporting each other in reflecting and improving rather than accomplishing external tasks, with modules for different professional skills including critical GenAI use. I will design and evaluate these groups as an adoptable drop-in assignment, as a separate one credit course, and when more deeply embedded in a computing ethics course.

Task 2) Video-assisted Comparative Reflection and Experimentation for Critical GenAI use: Building on a separate successful pilot, I will iteratively design, implement, and evaluate a novel "watch the video replay" video-assisted reflection, where students will record and analyze their process, such as when programming, both without and with using generative AI, then compare them, developing metacognitive awareness of programming and GenAI use. I will investigate how comparative reflection helps students understand when, why, and how to use GenAI critically. I will then design a refined longitudinal series of comparative reflections as students experiment with different ways of using GenAI, while reflecting on critical use. I will also create a module for augmenting that with group reflection, via Task 1's reflection groups. As a stretch goal, I may

investigate how to support students to transfer that learning about reflecting on GenAI use to co-designing GenAI agents and interfaces that support user reflection on critical GenAI use.

Task 3) Tools for Group Matching and Transfer of Professional Skills Learning: I will design and evaluate tools to sustain reflection group experiences and learning around professional skills and critical GenAI use, investigating support for sustained behavior change, lifelong learning, and transferrable skill application after coursework. This includes developing matching tools for ongoing reflection groups (including after graduation), and reflection group designs that integrate industry professionals so students and professionals both benefit from participation. I will also create a lightweight and more authentic technical environment to support student GenAI projects, enabling them to build and deploy agents that use the standard Model Context Protocol (MCP) to integrate with other applications. As a stretch goal, I may extend comparative video-assisted reflection and group reflection practices into professional computing and software development contexts. Figure 1 below shows the proposed research tasks, evaluation, broader impacts, and education plan.

	Year 1	Year 2	Year 3	Year 4	Year 5
Task 1	Reflection groups integrated in ethics course			_____ adapt further for broader adoption & potential stretch goals _____	
	"Drop-in" assignment reflection groups		One credit course reflection groups		
Task 2	Longitudinal comparative video reflection		Add self-experimentation		Add reflection on newer GenAI tools
Task 3	Tool for reflection groups matching and sustained behavior change			Design iteration and adding features	
	Pilot post-graduation professional reflection groups		Integrate alumni and professionals into reflection groups		
	Tools for mock workplace for GenAI agent development				
Evaluation	Iterative mixed-methods design evaluation				
	Pilot longitudinal evaluation		Scaled-up longitudinal evaluation		
	Mixed-methods comparative and faculty adoption evaluation				
	Qualitative empirical study of GenAI use via reflections & videos, surveys				
Broader Impacts	Increasing adoption of reflection groups and reflection on critical GenAI use				
	_____ Improving reflection, lifelong learning, and relationships in computing community _____				
Education Plan	Community group for students struggling with enabling skills				
	Hackathons and community events on integrating critical GenAI use into education				
	Reflection groups high school outreach program				
	200 level course on professional skills and GenAI use (3 credits)				

Figure 1: Summary of proposed research tasks, evaluation, broader impacts, and education plan.

2 Background and Related Work

2.1 Professional skills in computing education

In computing education there is a lack of scalable, adoptable, and effective ways to learn professional skills. Reflection, iterative improvement, and supporting others are essential professional skills for computing, for lifelong learning and ethical practice [90]. These professional skills and dispositions are a part of ACM Computing Curriculum 2020 [31] and also found in empirical research on industry [39, 40, 52].

In prior work in computing education, professional skills are often designed in a course to be learned in team projects trying to build software or produce some output [52, 110]. However, this is insufficient as there is a persistent computing academia-industry gap on learning professional skills; review and aggregate studies found continuous and lifelong learning [52, 56], and oral and written communication [112] as top skill deficits. A large 2025 ITiCSE working group found "...lifelong learning and professionalism are under-emphasised." [52]. Valstar et al found computing faculty think more resources are needed to teach and give feedback on professional skills [135], but faculty are chronically under-resourced.

For teaching and learning professional skills in computing in general, courses on specific professional skills [93], mostly communication [86], have been taught. For example, a course on communication skills focused on writing, speaking, and interpersonal communication might include modeling how to communicate your perceptions and interpretations while expressing empathy [18]. Approaches for broad curriculum integration exist but are rarely adopted [10, 36, 60, 61].

2.1.1 Reflection groups for professional skills

While reflection groups are effective and relatively common in teacher education and professional medical education [105, 137, 143], my work will adapt them for learning professional skills in computing. In computing education, there is no published work on more scalable regular non-project team reflection groups in general, nor designs for scaffolding peer learning in those settings with minimal instructor feedback. Recent literature reviews on group reflection find it is generally rare in practice and in research studies outside of those fields [5, 32, 147]. Teacher reflection groups related to computing exist such as [115], reviewed in [17]. Beyond computing education, a recent systematic literature review on group reflection research across disciplines found a gap for qualitative research on how members perceive the group's affordances, which our mixed methods longitudinal evaluations will address [147].

Some computing education work tries to teach reflection more or implement processes that include more reflection, but not via more scalable non-project team reflection groups where peer observation and feedback help drive learning. For example, some courses use a studio approach with a high staff to student ratio where teachers explicitly model and guide reflective critique sessions [25, 34, 111]. Babb et al present guidelines for integrating reflection [14]. Some courses integrate reflection [37, 45, 140], such as Burden et al who designed software engineering (SWE) process reflections within a SWE course [27]. In computing ethics education, some courses have consistent class breakout discussion groups [6, 88]. A few longitudinal qualitative studies include computing student reflections [108], and Barr, Andrei, and others found reflection skills took years to develop even with workplace-based learning [11, 15].

Some computing education work has facilitated more supportive spaces for learning and feedback, such as peer-led team learning (PLTL) or study groups aiding students with their homework, but not for learning general reflection and iterative improvement skills. PLTL involves trained peer facilitators or "near-peer mentors" not taking the course that act in ways similar to TAs [41, 50, 51, 126, 145]. Other work addresses various scheduling and other barriers to student participation in study groups, but not designed or evaluated for learning reflection and iterative improvement skills [29, 72].

2.2 Reflection in computing education

Published prior work on computing students reflecting on their programming has generally positive findings. As of submitting this proposal, in reviewing the literature we have found no published work on designs for

assignments for students reflecting by comparing programming sessions in general, nor comparing their process when not using GenAI versus when using GenAI.

Reflection [84] is common in volunteer and service-learning courses [34, 114] and computing ethics courses, mostly individual reflection [8, 24, 85, 131], such as logs or journals [33, 125] and self-evaluation with respect to ethical guidelines [35]. Peer feedback on assignments [110] and group class discussion of ethics is common and often about applying ethical frameworks to cases; for a broader review see [24].

Much of the work in computing education focuses on student reflection about comparing, evaluating, or explaining solutions to problems, rather than on reflecting on a problem solving process or programming session itself. This is often because the reflection is designed to accomplish worthy technical knowledge or narrow skill learning objectives; for example, for the learning objective “utilize the suitable map and reduce operations in Spark to transform and aggregate data” the reflection “Think of the solution you were attempting. Is this the same as what the bot presented in the chat? Take this time to explain the solution to each other” where the bot presented a solution that met the learning objectives [117, 119]. While that system involved a conversational agent to give hints and scaffold reflection of that kind, students did not reflect on their overall programming process, experience, or on about using the agent itself [94, 118, 119]. Similarly, some work includes collaborative reflection and students comparing, extending, and challenging other students’ reasoning [4, 119] or pedagogical code review finding issues with code itself, not the process of coding [64]. Other work tries to scaffold metacognitive reflection at a more tactical level such as while solving a particular kind of problem [46, 67].

While some excellent work closer to ours has students reflect on their GenAI use [54, 81], only a few works have reflection on problem-solving process and GenAI use, but none do comparative or longitudinal reflection. Some work has students reflect not on the programming process but instead comparing solutions, such as comparing one’s code vs. code solely done by generative AI [46]: e.g. “Compare the solutions and comment on the similarities and differences. Submit both solutions and your critique of the main differences between them.” The closest work is not an assignment but research on understanding students; it did a brief content analysis of student responses to an optional reflection question after an assignment on describing their process to complete the assignment, when and why they used AI, and how it was helpful or not [83]. Lyu et al analyzed some student open-ended reflections, and their course included using GenAI, pairing with a human, or both, but there is no description of the reflection questions or topics [81]. Long before GenAI but also in response to GenAI, work has called for increasing integration of learning metacognition in computing, including “critical thinking and reflection” [28, 46, 80, 133].

2.3 Video-assisted reflection for learning

While video reviewing is common in qualitative research and for coaching [49, 77, 79, 136, 149], having students review videos of their work as an assignment is rare in computing education, and comparing sessions appears novel. Students have recorded retrospective video explanations of their solutions, such as on assignments [139] or as “post-exam videos” [138], like an oral exam where they explain and correct mistakes as they explain their solutions. Some works study video reflection for teacher training [7, 23, 102].

There is work about reflecting on code editor replays (keystroke level) on small programming problems, but not videos which also show more information seeking and other behaviors on the screen. We are aware of only two works. In 2021 Ditton et al piloted editor replays finding some benefits and student frustrations around the value of reviewing them [38], and in 2023 Xie et al found students who reflected on code replays exhibited better self-regulation like planning more often [146]. There are also research studies where the researchers analyzed videos of student programming sessions while using generative AI [109], which involves students thinking out loud, or reflecting in a diary or survey question [83], or during an interview [92], but not as an assignment. Just as video helps researchers analyze interactions in detail, video-assisted reflection may help learners notice good tactics and other patterns, and also reduce cognitive load [91] and recall bias [76, 127, 130], which seems likely to aid reflection.

2.4 Understanding learning of critical GenAI use skills and adoption of GenAI

There are many surveys of student and other user perceptions, and evaluations of various learning interventions and GenAI systems for aiding learning [9, 19, 20, 63, 66, 70, 74, 103, 106, 107, 116, 123, 148, 150]. However, there are very few longitudinal mixed-method evaluations [107], such as [70, 83, 124], and we are unaware of any "in the wild" empirical studies of users doing video-assisted reflection on their generative AI use, much less comparative reflection. By creating reflection designs and studying their reflections, we will also study how students think about critical AI use in a deep and rich way longitudinally. The closest work appears to be the PI's prior work qualitatively interviewing and analyzing change over time in a course with deep generative AI integration [121, 124]. If we are successful in having professionals volunteer to do their own video-assisted reflection and share either their reflections or their reflections and videos with us, that will also be a novel "in the wild" dataset on using GenAI and reflecting on critical AI use. Separately, in following up on students, we will also ask them how their reflection has influenced their GenAI use and their ethical thinking, which will be another less novel but meaningful contribution.

2.5 Tools for matching people for learning and reflection

Our work generally departs from prior tools for facilitating professional mentorship [21, 57, 113, 122, 129], as our reflection groups are designed for professionals and students to learn and reflect as peers [62] rather than as mentors (encouraged by our preliminary work on reflection groups where professional MS students reported benefiting similarly, if not more deeply, compared to undergraduates [95]). We will build a tool integrating matching and follow up evaluation for reflection groups, with mostly non-novel community discussion, group matching for similarity and scheduling using constraint solvers, and basic behavior change features leveraging nudges [30, 141], additional reflection prompts, and accountability features [48]. This tool will have a novel generative AI based "share anonymously" feature for reflections, and a feature to transform private reflections in reflection worksheets into a blog post format [59] people can share, to encourage students and professionals to share their reflections on GenAI. This may lower the barrier to public sharing, and encourage broader discourse about GenAI within student and professional communities.

3 Research Plan

Each task describes our preliminary work then proposed work. All preliminary work already has IRB approval, which was written broadly such that it also already covers the proposed work in Task 1 and 2. Figure 1 on page three summarizes the proposed work as a whole.

3.1 Task 1: Reflection Groups for Professional Skills Development

3.1.1 Preliminary Work

The reflection group design carefully scaffolds student peer observation and peer learning, via regular outside-of-class weekly reflection meetings with agendas and goal-setting [95]. We provided scaffolding for the group meetings via a weekly reflection meeting agenda. During group meetings members mixed writing in the agenda and discussion, an idea inspired by prior work [68]. After each meeting, students turned in individual weekly reflections graded for submission; attendance was not directly graded. We implemented and studied these reflection groups in a three credit undergraduate technology ethics course, taught synchronously online only. We also frequently used these groups during class periods for active learning activities. We received IRB approval for this and all other preliminary work.

Reflection groups were created using survey data in the third week that asked students about scheduling constraints and identity through shared gender and disability status. There were a total of nine reflection groups for 38 students. This gave one all disabled women group of four, one mixed gender group of 3, one mixed group of five with 3 men (an evening all professional MS students group). The remaining five all men groups included one all disabled men group of 4, two groups with two of four members disabled, and a group of 4 with one disabled. No gender or disability status was disclosed by the instructor or TA to students, only group numbers and meeting times.

As part of the reflection groups assignment, each group had a ~30 minute weekly reflection meeting online outside of class time. Each reflection meeting was broken into three different parts. First, students would share their progress and challenges. Next, they would affirm challenges encountered, encourage each other, and celebrate accomplishments by commenting on progress and perseverance. Last, students would generate ideas for improvements. Throughout the meetings, students could potentially offer help to group members. For the weekly reflection meetings, students were provided a meeting agenda with guided questions to help support conversation, particularly to scaffold their first meeting.

Before the group's first weekly meeting, to form critical initial group bonds and safety, we adapted an activity from experimental psychology for creating interpersonal closeness among two people [12], as an icebreaker. We adapted it by removing questions that seemed might evoke trauma—such as “If you could change anything about the way you were raised, what would it be?”—and, for the group setting, shortening the duration allocated to questions, for example changing from *four minutes* to “Take a minute and tell your teammates your life story in as much detail as possible.”

We conducted a qualitative thematic analysis [22] of end-of-semester group reflections, performed by the PI, two students from the course, two students not in the course, and an external qualitative computing education researcher [95]. Each theme (similar to a category) required supporting quotes from at least three students and most included quotes from more. The themes for student reported learning ranged from learning skills, often professional skills (such as communication), to skills and dispositions for improving their ability to improve (such as being more honest with themselves and new habits of reflection, experimentation, and smaller more reachable goals). Some students even transferred their learning and reported positive impacts in their relationships outside the course, even in their families. The themes for student reflections on what helped their teams work included a sense of community and shared experiences, and growing trust and safety to be vulnerable. This led to more openness and honesty, which enabled feedback loops including sharing goals, problems, solutions, and advice. Students also mentioned basic features of the reflection groups assignment, such as the weekly group meeting agenda and submitting post-meeting individual reflection question responses. The professional MS students also benefited, even when in groups of undergraduates.

3.1.2 General Approach to Designing Reflection Groups for Adoption

We will conduct design-based research [43], cyclically following the steps of 1) design, 2) deploy, 3) evaluate what worked and did not, and create theory to understand why. Then, we will refine our design and repeat the process. Initially, we expect the design to change more, then stabilize over time. With each design iteration, we will incorporate ongoing longitudinal evaluation and compare the results from the previous designs. The mixed-methods longitudinal evaluation design is shared across sub-tasks and is detailed later in 3.1.5.

Throughout this process, we will invite students and faculty to co-design workshops, until the design stabilizes, which we expect by end of Year 2 or 3. For the “drop-in” design, we will invite them to outside of class workshops. For the deeper integration courses, we will include the co-design workshop in a class session in the course. For all the courses, the end-of-semester retrospective reflections will also have a debrief component about what might be improved and ideation about changes to the design. These will also be integrated into our ongoing qualitative analysis and evaluation.

We will also include a yearly design cycle for adoption with faculty, including a survey on interest and perceived barriers and benefits and optional interviews, and a design workshop where we invite faculty who did integrate, attempted to integrate but encountered barriers, or chose not to try to integrate. This will involve understanding faculty and their perceptions, and co-designing how they might use and/or adapt the design for their courses and institution. From our preliminary work already, faculty in and outside UMaine (for example, see Dr. Velasco's letter of collaboration from Duke University) have already expressed interest in attending virtual design workshops and potential adoption of reflection groups. The design workshops will also allow attendees who are not the PI to “spin off” a version of the design and evaluate it, where we will strongly encourage them to use a minimal pre- and post-test quantitative evaluation. The PI will also hold by appointment “office hours” for interested faculty, while also taking notes and asking optionally for consent to include them in research on designing for adoption. Once each subtask has a reasonably successful design

with positive pre-post outcome data and qualitative results, the PI will advertise and recruit faculty more broadly for those.

3.1.3 Subtask: Design and iteration of deeper course integration for ABET tech ethics course

Our approach mitigates risk by building on the ethics course that deeply integrated the reflection groups, including frequent use of the same groups in class activities and discussions. This version will effectively "lead" the drop-in and one credit versions by a year, and also inform design for those versions.

First, in Year 1, we will deepen the evaluation step of our preliminary work with follow-up interviewing and surveying the students from that course and also other students who did not take that course for comparison. We will focus on their experiences transferring and maintaining the professional skills they learned, and exploring what effects the groups had. While this will not yield any causal differences, it will help us understand barriers to maintaining and transferring professional skills they learned to inform the next iteration of the design. This helps us conduct early tests and refine interview protocols for future longitudinal evaluation. We will also test a small set of quantitative outcome measures and find new or better ones as needed.

Based on those results, we will then do a design iteration to create a new version, improving *long-term transfer of professional skills* (students maintaining, and actually using, those skills outside and after the course). We will evaluate this version with a pre-post mixed methods design including interviews and quantitative measures, and also longitudinally on the same follow-up timescale to compare with version 1 (our preliminary work), aligned to have the same duration. Importantly, our institution has another tech ethics course in the other semester; we will recruit and survey in the other instructor's course at the beginning and end of the course, and longitudinally. We will do the same for the intervention group. Depending on the distribution of the data, we will attempt to use a matched case-control design for a comparative quantitative evaluation at different time points. As a fallback, we will compare within subjects pre-post changes and pre to longitudinal post changes.

In Year 2 and 3, we will do more design iteration, focused on transfer and faculty adoption. In Year 4 and 5 (or sooner if successful earlier), we will shift our focus to designing for adoption by faculty at a broader set of institutions. Faculty will likely adapt parts of the course as they adopt it, and we will ask faculty to use the short survey outcome measures as pre and post tests so we can compare variations in that. We will also recruit other ethics course offerings at their university(ies) for comparison groups.

In Year 4 and 5, we will facilitate and study adoption quantitatively and qualitatively. The PI will assist faculty via the PI's faculty "office hours" (limited to 1-2 hours per week) in customizing the course, and will, with proper consent procedures, take notes and analyze that as qualitative research. We will also discover the kinds of changes faculty think are good but are not helpful, and changes the PI thinks are not helpful but are actually helpful, and facilitate design dialogue and a community of practice around that in the yearly faculty design workshops for reflection groups. We will also include at those workshops creating notes and helpful tips and materials for presenting the results of the research and student outcomes to aid adoption, as well as a guide to discussing them with other faculty. Based on all of those, we will refine the designs, including producing a set of modules on professional skills and ethics topics to encourage adoption.

3.1.4 Subtask: Design and iteration of "drop-in" assignment and 1 credit course version of reflection groups

Based on lessons and designs from the deeper course integration, we will do design iterations to make reflection groups work as a drop-in assignment with the only integration being using a class session for the icebreaker for forming psychological safety and initial group connection. However, we expect to encounter some challenges as a drop-in assignment as, in the ethics class, having the groups also discuss ethical topics in class may have helped strengthen ties and make the groups more vulnerable. To address this, our Year 1 design iteration will focus on creating modules, which effectively create a more careful sequence of varied meeting agendas (each week's agenda with particular reflection questions is a module) for particular professional skills, to scaffold gradually opening up in the group. For example, a simple discuss your career

goals, concerns, and what you can do module was noted as particularly useful by students in our preliminary work; this is an example of a module for the meeting on a specific topic. Among the modules, we will also include two to three modules on critical GenAI use. Each module will also include one or two questions from the icebreaker activity to help reconnect and set the tone of sharing openly.

We plan to do this in at least two different courses: a three credit introductory creative media and computing course, and a one credit first year CS majors required orientation seminar on basic college skills and the computer science department and major (see Ippolito's then Dufour's letter of collaboration). Beyond the addition of modules, we will create these versions by initially keeping the same design as in the ethics course, but without using the reflection groups also as groups for active learning during the normal class sessions.

At the end of Year 1, we will evaluate and use our results and student and faculty feedback, and from our qualitative interviews and other evaluation data. For example, if we don't have similar positive results as the preliminary work, this may indicate having the reflection groups also do active learning activities during the normal course class activities is necessary for the groups to work well. If so, in Year 2 we will add that back into the design. We chose not to include this in Year 1 because, in preliminary discussions with faculty around adoption, we observed a common negative reaction to the idea of reducing mixing of students in class. We anticipate our design work for this task area in general will be doing this kind of balancing around designing for adoption, and we think it is where we will find more challenging and potentially hard trade-offs. We will draw from work on reflection groups from medicine in addressing these and other trade-offs [82].

In Year 2 and 3 we will do design iteration for that and other trade-offs. For example, in the current design of the groups, we did not observe a group that was negatively affecting its members. Instead it was reported as moderately positive or not affecting them. Still, we should expect some groups will not work well, so we will come up with a remedial mechanism, allowing the students to choose to 1) join a more functional group (which opts in to adding one member), 2) join a new group with others. We believe our design not grading directly on group attendance makes the groups more resilient as a whole, where students who are not ready or really do not think the groups help them can just choose to not participate. While we have not observed a group with negative effects yet, we will also talk with those students to co-design with them how to respond if that situation arises.

If the "drop-in" reflection groups are particularly successful and become more widely adopted, we will experiment with a "one reflection group across all courses" model. We will house the small grade incentive for participation in particular required courses, or with the groups hosted in a separate one credit course. We will also explore inviting students to switch groups when the semester changes, versus keeping same groups, although we will not force students to change.

With each design iteration we will incorporate ongoing longitudinal evaluation and compare the results of previous designs. During the course we will do formative evaluation of the reflection groups by observing agenda notes and individual reflection submissions, collecting student feedback, and analyzing student feedback to guide real-time adjustments and inform future improvements.

3.1.5 Evaluation Plan

We will conduct mixed methods longitudinal evaluation. For the quantitative outcome measures, to increase response rates, the longitudinal follow-up surveys will have a relatively small set of measures, all of which will also be in pre and post course surveys [134]. It is challenging to find domain specific and short professional skills measures. We will start with the 20 question Self-reflection and Insight scale [55], a recent 25 item self-directed learning scale intended for measuring lifelong learning skills [78], a 7 item psychological safety scale [44], and a high quality one question STEM professional identity measure with a strong validity argument [87]. In Year 1 we will pilot on the pre and post course surveys a wider variety of measures, and choose more to include on future surveys (or replace ones on the small set) based on observed sensitivity and domain relevance.

We will evaluate student individual and agenda reflections using two qualitative rubrics and a qualitative thematic analysis. The first rubric is a simple three category rubric [69] on each student as *non-reflector* (i.e.,

lack evidence of deliberate appraisal), *reflector* (i.e., demonstrate insight through analysis, discrimination, and evaluation), or *transformative reflector* (i.e., indicate a transformation from initial perspective); we will compare change over time and when transitions happen. We will also use a reflection rubric from medical education [89]. We will also conduct our own more domain-specific qualitative thematic analysis [22]. In doing qualitative analysis we will follow practices for rigor, such as reporting sources of disagreement and related analysis discussions [58].

We will also evaluate benefits via qualitative analysis of longitudinal interviews on professional skills behaviors and challenges at work, self-reported objective career events like obtaining internships, interviews, and jobs, and proximate measures like people choosing to continue participating in reflection groups and degree completion.

3.1.6 Risks

Supporting groups to have psychological safety and also reflect well is challenging. We mitigate this risk by both refining the ethics version that worked well in preliminary work, and iterative design for the "drop-in assignment" and one credit course versions.

Assessing reflection skills directly is hard and also cannot be used as a proxy for transferring professional skills [73]. To mitigate this, we will also longitudinally measure professional skills behaviors via surveys, and with a smaller set of qualitative interviews where we ask graduated students/professionals to describe specific instances and behaviors and show relevant work-products.

Participant non-response over time in longitudinal research is an issue, but can be mitigated with techniques like community building (such as connecting with students by sincerely thanking them, and framing the follow-up as wanting to understand them (which is true)), following up regularly, and making responding easier; compound strategies like this are common in studies with $\geq 80\%$ retention rate [3, 134]. For quantitative data we will also test for response bias and report limitations if found, or apply corrective statistical modeling as appropriate.

Training students in qualitative interviewing is a risk. The PI has encountered common pitfalls of new graduate and undergraduate students interviewing, and has learned to teach and monitor these skills closely. We will practice at least two full mock interviews before each round with research lab members.

3.2 T2: Video-assisted Comparative Reflection and Experimentation for Critical GenAI use

We will use design-based research to iteratively complete qualitative study of student GenAI use and evaluate students and student reflections (subtask 1), and then use that to refine the design of longitudinal comparative video reflection assignments, including students experimenting with different ways of using GenAI (subtask 2).

3.2.1 Preliminary Work

We have created a novel initial design for video-assisted comparative reflection [47]. To do this, we applied the DEAL framework, which is a high level architectural pattern for designing reflection. The sequential steps in DEAL are Describe, Examine, and Articulate Learning [13]. Normally each step has a set of reflection questions to respond to, usually in writing.

The Describe step was to a) work on their team software project while recording a video of a session without using GenAI, then a session with using GenAI, then b) later review each video by watching it. In the Examine step, students answered a set of written reflection questions in the following order. To answer these, the assignment had students fill in a table with four columns with headers "Reflection question", "Answer for Session 1", "Answer for Session 2", "Answer for Compare the differences". Each row had one reflection question, for example, "What were you working on?". First, students were instructed to watch video 1 then answer all of the Examine questions, then repeat that for video 2 (we will describe the "compare the differences" column in the next paragraph). The sequence of questions started with questions to make sense of what happened at a higher level: 1) What were you working on? 2) What progress was made? 3) If you did pair programming, who did you pair with?, 4) What resources, people, or tools did you work with?

5) What challenges did you encounter?, and 6) How did you respond to each challenge?. Next, students answered detailed questions on roughly what percent of their time was spent in different time categories while programming, corresponding to basic steps in the programming process. The first three time categories were: planning, writing or editing code manually, debugging or understanding code. Next were three time categories related to information seeking and human collaboration: looking up and using examples, looking up and using API documentation, asking another person for help, and working with another person. Finally, there were time categories for using GenAI: using GenAI to write code, using GenAI to understand code or ask questions about APIs, etc., using GenAI to debug code, using GenAI for planning.

In the Examine step, students also compared their two programming sessions, without generative AI, and with generative AI. First, they answered what could be improved about each session, and what they did differently for the second session (using GenAI was usually the only change between sessions). For each reflection question from the prior steps, students answered “Compare and Evaluate: write about 3-5 bullet points total per question on a) the differences between the two videos / programming sessions, b) what caused those differences.” Those answers were written in the last column of the reflection questions table, each in a cell with “differences:”, some whitespace, then “causes of differences:”.

Next, for the Articulate Learning step, students answered the following questions. For context, video 1 was the session without using generative AI, and video 2 was the session with using generative AI. The last five questions are from the DEAL framework examples in [13], with the last question broadened:

1. Summarize the most significant differences between your software development session in Video 1 vs. Video 2.
2. Summarize the causes of those differences and what their effects/consequences were.
3. What did I learn?
4. How, specifically, did I learn it?
5. Why does this learning matter, why is it important?
6. In what ways will I use this learning?
7. What goals might I set in accordance with what I have learned in order to improve myself and/or the quality of my learning and/or the quality of my future?

For our analysis, we conducted a qualitative thematic analysis [22] of the student’s written reflection submissions [47]. Our analysis team was a person unaffiliated with the university and the course instructor (the PI). We first wrote memos on students’ reflections answers, reading each student as a whole, which we then discussed. We then decided to, based on the bullet point format used by students, to scope our analysis to specific questions to make a large affinity diagram of all of the responses, clustering them to then create themes/categories. We chose questions from the “articulate learning step” listed above, #3, 6, and 7. We chose these questions to balance a manageable set of data to analyze while also still covering what students learned, as well as how students may use their learning and the future goals they might set, which helps gauge supporting their reflection to be actionable for putting that learning into practice.

We found that this reflection assignment supported students’ learning about generative AI, including its downsides, negative impacts, and limitations, while also supporting reflecting on more critical usage patterns and prompt engineering skills [47]. Students also demonstrated learning about programming in general, such as the need for background understanding before coding, planning, human help-seeking, meta-cognitive insights about their learning processes, and even professional identity development. Students in the reflection generated very diverse goals for acting on their learning, including more skillful and critical GenAI use, but just as often on improving programming generally with better help-seeking, human connection and collaboration, and even purposefully not coding with GenAI.

3.2.2 Deep Qualitative Analysis of Videos, Student Reflections, and GenAI use

First, we will interview students who did the video reflections to identify and understand barriers to completing and doing the goals they set from their reflections. We will also evaluate if they adopted regular reflection behaviors, and if so, what were the enablers, and if not, what were the barriers. In the interviews, we will end

by getting student feedback on the reflection assignment design and potential designs for improving that and extending it to include following up and reflecting over time. We will also analyze their videos and actual written reflections from the assignment, using similar methods as in Task 1. We will repeat this process for each design iteration of the comparative video-assisted reflections.

As a stretch goal, if they become available, we will also gather and include log data via students using open source GenAI IDE integrations, or ask them to consent to logging plugins for such tools or GenAI agent tools. There is a hard trade-off on students using the best tools, which may be closed like Windsurf, versus less good tools that have logging in them. We would rather have more ecological validity.

3.2.3 Design longitudinal video-assisted comparative reflection, including students experimenting with different ways of using GenAI

Using that evaluation, we will then design a refined longitudinal series of comparative reflections as students experiment with different ways of using GenAI, while reflecting on critical use. In Year 1 we will have students do the reflection three times in the semester, and add a midpoint and final metacognitive component, involving for example "reflect on patterns in your reflection" and identifying persistent blockers. We will also add using and reflecting on GenAI agents for coding at the end of the longitudinal series. We will do this in the PI's software engineering course and Ippolito's advanced programming courses (see letter of collaboration), and, once successful, recruit faculty from other institutions to adopt that reflection assignment.

In Year 2 and Year 3 we will add self-experimentation on AI use strategy, such as using GenAI only to aid learning, versus using GenAI for "vibe-coding" [120]. In Year 4 and Year 5 we will add reflection on not yet invented GenAI tools from industry or research.

3.2.4 Evaluation Plan

First, for each iteration of the video reflection assignment, we will do the qualitative evaluation from subtask 1 above. For quantitative evaluation, we will have pre and post surveys, demographics, regular during course surveys, and longitudinal post surveys, on students' generative AI use in and outside the course, and quantitative outcome measures, including the refined outcome measures from Task 1's pilot evaluation. We will add measures focusing on critical GenAI use such as the ChatGPT Literacy scale [75], and the PI's short four question critical and ethical GenAI use self-efficacy (i.e. confidence) measure [98].

From those students, we will also conduct longitudinal qualitative interviews with a purposeful subset of students. These will also include their changing GenAI use and critical use skills. We will sample stratified by gender, the Self Reflection and Insight scale [55], and generative AI critical and ethical use self-efficacy [98].

For comparison on critical GenAI use skills, we will also conduct interviews of professionals, for qualitative research and also designing larger quantitative comparative evaluation. We will separately recruit a comparison group of professionals to survey and then offer interviews to. Using the survey, as a nice to have, we will attempt to create a roughly matched comparison group to previous students who did the reflection assignments and have become professionals. The goal of this qualitative analysis and comparison is to understand differences in experience and potential effects, which we will use to identify and pilot relevant quantitative measures.

We will implement those measures in longitudinal quantitative surveys for more rigorous quantitative comparative evaluations, starting in Year 3. This will have much larger populations in the intervention (roughly 60-100) and comparison groups (roughly 200 recruited via a professional research participant recruiting and sampling company). We are planning for this study from Year 3 to end of Year 5. We will over-recruit participants anticipating a drop-out rate of 50%; we will adjust this as we observe it empirically. We will also implement best practices to reduce participant drop-out as described in Task 1's evaluation plan.

3.2.5 Risks

As in our preliminary work, we will instruct students to turn off notifications and follow other privacy increasing behaviors when recording their screen videos. The instructor will also give feedback on first video submission in the longitudinal series to increase data quality. We already have IRB approval for the student

studies, and we will, as always, follow normal IRB research ethics, but also go beyond as we allow students to opt-out of the research at any time, such as after they submit assignments and even after the course ends (as of now no opt-outs yet). We have also budgeted for a video anonymization platform to process and help automate part of reviewing and removing unintentional private information in the videos.

3.3 Task 3: Tools for Group Matching and Transfer of Professional Skills Learning

Building on the work in Task 1 and 2, we will build support tools for reflection groups and for student GenAI development, to aid long term use and transfer of professional skills and also support professionals to participate in reflection.

3.3.1 Preliminary Work

The PI has manually matched groups and schedules in the preliminary reflection groups work, but not designed or built automated tools for that, or frameworks for GenAI assignments yet. The PI has designed and piloted isolated basic GenAI agent building assignments. In preliminary work we observed professional MS students benefiting from their reflection groups, even when in groups with undergraduates [95].

3.3.2 Tool for reflection group matching and sustained behavior change, including professional participation in reflection groups

We will design and evaluate tools to sustain reflection group experiences and learning around professional skills and critical GenAI use, investigating support for sustained behavior change, lifelong learning, and transferrable skill application after coursework. This includes developing matching tools for ongoing reflection groups (including after graduation), and reflection group designs that integrate industry professionals so students and professionals both benefit from participation. To design these tools and groups, we will also draw on work about professional participation in reflection groups from medical education [26, 42]. To build these tools, we will use open source components, such as for scheduling [2].

To help gather requirements, in Year 1 we will invite students from the preliminary ethics course (mostly graduated) and other courses in future years, for ongoing reflection groups. Starting in Year 2, supported by our matching tool, we will also iteratively design student reflection groups that include professionals as peer members. In Year 3 and beyond, we will iterate on the tool, including features for sustaining behavior change, and other features suggested by users and from our ongoing evaluations; for example, reminders and nudges [30, 141], additional reflection prompts, and accountability features [48]. We will also build novel GenAI based features to a) "share anonymously" their reflections (reviewed by the user before posting), and b) a feature to transform private reflections in reflection worksheets into a blog post format [59] people can share, to encourage students and professionals to share their reflections on GenAI. This may lower the barrier to public sharing, and encourage broader discourse about GenAI within student and professional communities.

3.3.3 Tools for more authentic mock workplace infrastructure for student GenAI agent development

In Year 2, we will begin designing a more authentic mock workplace tech infrastructure for building and deploying GenAI agents, a tool for managing and creating those environments, and an example assignment that uses those. The PI will teach building a simple GenAI agent that uses the standard Model Context Protocol (MCP) to integrate with other applications; for example, the agent might be given a conversation transcript on user stories, then use MCP to create issues for them in the Github issue tracker for the project. The research team will build a tool for making a copy of open source or student Github projects and task management trackers, so students can work in a more authentic environment than just a made up environment with made up data. We will initially build and pilot this in the ethics course mentioned in Task 1, and the PI's software engineering course. Students will design and build ethical generative AI agents for professional environments, which include features for supporting user reflection and evaluating impacts of GenAI use, such as the agent asking their users to reflect occasionally or ask them what they think about potential ethical issues. The PI will create instructional and assignment materials for that, and part of the research is to figure out how to teach what an ethical agent might be. To enable deploying and evaluating those agents, we will

create a tool to copy a template model work environment based on free collaboration tools on Github, with example data in project management tools, codebases, and other tools, so students can practice using MCP and seeing issues around that, like an agent unintentionally deleting or destructively rewriting tasks in the issue tracker over MCP. As a stretch goal the copying of the template will include GenAI generating tasks and content (not code) for a workplace customized to the student, like a project at a local employer they desire to work for.

That “snapshot copy” infrastructure will enable experimentation and mitigate risks for using student-created agents, by also allowing copying any public Github repo and project task management repository. This includes student projects, such as from a SWE course, so they can try out agents in a safe environment that won’t disrupt actual student projects. This copy function will also allow creating regular backup checkpoints in case an agent is helpful enough that those SWE students might want to use it in their project; while crude, this helps mitigate risks (and any unintended failures would be important learning experiences for students).

We will also host infrastructure for running open source private LLMs, including the latest tool-using agents such as Kimi K2 [1, 53], with a server with 768GB of VRAM on 8x Nvidia RTX Pro 6000 Blackwell GPUs. We will also make this available to other computing education researchers and courses based on available capacity. This will allow us to study willingness and use of students using LLMs, when inputting data which they may otherwise hesitate to share with OpenAI, Anthropic, DeepSeek, etc or off-campus cloud-hosted “private” models through API use. This will also allow us to study/support the willingness of ethically concerned students to use GenAI, as we can host the servers without incremental evaporative water cooling environmental costs. Those costs lead some environmentally conscious students to not want to use GenAI or use it less [132, 142]. This is also sized to allow running high quality smaller quantized open source models for 100 concurrent users, such as when students are working on a project deadline.

3.3.4 Evaluation Plan

We will primarily use the mixed-methods evaluation designs from Task 1 and 2, while also combining them where appropriate. For Task 3 subtask 1 on the reflection groups tool and professional integration, we will primarily use the reflection groups (Task 1’s) evaluation design, with pre and post course surveys replaced with pre group initiation, 3 months later and 6 months later and further longitudinal surveys. That will also include interviewing professionals in the groups, and recruiting for a comparison group via a professional research subjects recruiting company. For Task 3 subtask 2 related to developing and using GenAI, we will use the critical GenAI use (Task 2’s) evaluation design. The PI will also use their expertise in educational assessment and validity [96, 99] to augment those with any newly published relevant high quality measurement tools with validity arguments.

3.3.5 Risks

Professionals may want to just do reflection groups with other professionals. If so, professionals doing group reflection is still a positive outcome, but we will also iterate on that as part of our design process. We will also facilitate matching alumni to students from their university. IRB approval is not a risk as voluntary participation by professional participants is normal, and students will not be required to have professionals in their group, they will need to opt-in after reviewing the professional’s profile.

4 Education and Outreach Plan

The proposed education plan outlines projects for high school, university students, and the community. These are in addition to the proposed research activities, and will not be conducted as research activities for this proposal.

Context and Commitment. As a faculty member with ADHD, IBS, autoimmune, and chronic health issues, I am truly committed to mentoring all students, especially those that struggle with time management, executive function, anxiety, and/or health issues. I will refer to these as challenges with *enabling skills*. In general and at UMaine this is relatively common, including students with diagnosed issues; for example, data from

UMaine Student Services shows approximately 25% of computer science students are diagnosed with autism, and in my courses in surveys students self-report around 40% related to challenges with *enabling skills*.

Activity #1 Bi-weekly community group for students struggling with enabling skills. Last year I started such a group, with roughly ten participating undergraduate and graduate students. I will continue this community group, and add optional recurring feedback and evaluation surveys to iteratively improve it, including each semester.

Activity #2 Hackathons and community events for conversations on integrating critical and ethical GenAI use into education Separate from the proposed research, I will host virtual community events to facilitate students, faculty, and community members across institutions to discuss and co-design integrating GenAI into education. I will also organize and facilitate a one week virtual hackathon each year in the summer on a) adapting and open-sourcing participants' existing curricular materials and b) building new ones, including GenAI prompts and agents for education, focused on addressing challenges with *enabling skills*.

Activity #3 Reflection groups high school outreach program. I will create a new outreach program inviting high school students to virtually attend and/or participate in reflection groups with undergraduate student mentors in the higher performing reflection groups. This can increase interest in college and computing.

Graduate and Undergraduate Classes. The courses I teach directly overlap with my proposed computing education research, including reflection assignments in my software engineering and my ethics course. That includes heavily refining my technology ethics course including integrating GenAI, while also meeting ABET requirements. Beyond that, I will also develop a new three credit 200 level elective on professional skills for lifelong learning and generative AI use, building on and integrating my research. I will also monitor and incorporate into my courses the latest research on teaching writing with GenAI.

5 Broader Impacts

By developing scalable reflection-based pedagogies, the proposed research will improve computing students' professional readiness, ethics, and ability to manage and co-lead technological change. The focus on psychological safety and vulnerability in learning helps make everyone more successful in computing education, while the emphasis on critical AI use prepares students to be responsible leaders and co-designers of GenAI integration. Open-source curricular materials and faculty development will enable widespread adoption across rural and urban computing programs. These will support and help improve computing graduates' employment outcomes. Stronger, more vulnerable human relationships and professional networks will contribute to positive cultural change for more critical and ethical computing in general and for GenAI, across education and industry.

6 Intellectual Merit

The proposed work will fill the gap in scalable, adoptable, and effective techniques for learning professional skills in computing education, including lifelong learning and critical and reflective GenAI use. It will advance knowledge in computing education research (CER) and human-computer interaction (HCI); this includes novel designs for and longitudinal mixed-methods empirical evaluations advancing our understanding for: 1) teaching and reflection methods for learning critical and ethical GenAI use, 2) longitudinal video-assisted comparative reflection by learners inspired by qualitative research techniques, 3) empirical understanding of GenAI use, including critical reflection on use, learning over time, and how, when, and why students do and do not use GenAI, 4) supporting group reflection in synchronous online settings, and 5) bridging student and professional learning communities to reflect and improve together.

7 Results From Prior NSF Support

The PI has not previously served as a PI or Co-PI on a funded NSF proposal.

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